Manideepika Reddy Myaka

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EDUCATION

Master of Science in Computer Science - North Carolina State University **August 2023 - May 2025**
Bachelor of Engineering in Electrical Engineering - Chaitanya Bharathi Institute of Technology **June 2018 - June 2022**

SKILLS

Certifications: Google Cloud Certified **Associate Cloud Engineer**, NVIDIA Certified Professional: **Agentic AI**
Open Source Contribution: BugsinDLLS, Centaur, Tensorflow, Pytorch, Jax
Languages: Python, Java, C, JavaScript, SQL
Generative AI: LangGraph, MCP, RAG, LangChain, OpenAI, vLLM, Hugging Face, FastAPI, AutoGen
ML & Databases: Scikit-Learn, TensorFlow, Keras, Jax, PyTorch, PostgreSQL, MongoDB, Pinecone, Redis, Neo4j
MLOps: AWS EC2, GCP, Vertex AI, Docker, Kubernetes, Apache Kafka, Git, Terraform

PROFESSIONAL EXPERIENCE

Research Assistant (SWAT Lab) | North Carolina State University **July 2025 - June 2026**
Deep Learning Compiler Validation Platform

- Built a differential-testing framework for deep learning compilers leveraging **Gemini** to generate diverse deep learning models and identify compiler inconsistencies across execution backends.
- Identified and reported **76 previously unknown issues** across **PyTorch** (37), **TensorFlow** (24), and **JAX** (15), contributing to improvements in compiler correctness, reliability and framework robustness.

RBFuzzer: Rule Based Testing Framework for Deep Learning Systems

- Conducted large-scale analysis of **283** real-world **deep learning api** failures, to characterize recurring fault patterns and root causes to derive actionable testing strategies.
- Developed a Fuzzer, an **LLM-guided** Python testing framework that converts root-cause insights into adaptive input mutation strategies for automated test generation and execution.
- Automated failure reproduction and triage workflows, resulting in the **discovery of 50 bugs** in **TensorFlow** and **PyTorch** and supporting collaboration with open-source maintainers through issue resolution.

Software Engineer | Cognizant **January 2022 - June 2023**

- Developed and maintained backend services and **API validation** frameworks for enterprise-scale web applications, improving release stability and overall software quality.
- Designed automated testing and monitoring pipelines using **Selenium**, **TestNG**, **REST APIs**, and **Postman** to strengthen regression coverage, reducing production defects by **95%**.
- Automated CI/CD workflows with **Jenkins** and **GitHub Actions** and optimized **SQL queries** and database processes, **reducing deployment times by 40%** and **improving application performance**.

PROJECT EXPERIENCE

Ellipsis: Podcast Generation Tool | *React, Redis, llama.cpp* **May 2025 - June 2025**

- Built an end-to-end **LLM-powered** podcast generation platform using **React.js** that converts user prompts into structured multi-speaker scripts and synthesized audio, reducing manual scripting effort by 70%.
- Optimized inference performance using **llama.cpp** and streaming token generation, achieving 1.2-1.8s initial token latency and ~85% lower projected costs compared to hosted API-based scaling models.
- Designed a **multi-agent** content refinement pipeline with **Redis Pub/Sub + SSE** streaming and modular multi-speaker TTS, improving script coherence by 50%, and minimizing factual inconsistencies by 60%.

Online Book Shopping System | *Ruby on Rails, PostgreSQL, HTML, CSS,* **September 2024 - October 2024**

- Developed a full-stack e-commerce platform using **Ruby on Rails (MVC)** and **PostgreSQL**, implementing user authentication, book catalog management, shopping cart functionality, and order processing.
- Containerized and deployed the application using **Docker**, **Kubernetes**, and **AWS (EC2/Lambda)**, enabling scalable service orchestration and automated infrastructure management.
- Implemented **OAuth 2.0 authentication** and built comprehensive **RSpec** unit and controller test suites, achieving 90%+ test coverage across validations and CRUD workflows.